

**Cool summers are stormy summers: precession reorganizes extratropical
cyclone activity through the quiet season**

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7 ABSTRACT: Extratropical cyclones peak in winter, while orbital precession reshapes the sea-
8 sonal insolation cycle most strongly in summer. Thus storm tracks might be expected to ignore the
9 precession cycle. This is examined with twelve equilibrium simulations using a coupled climate
10 model AWI-ESM2, spanning a full precession cycle at 30° longitude increments under fixed eccen-
11 tricity and obliquity. Extratropical cyclones are explicitly tracked based on 6-hourly atmospheric
12 pressure fields for both hemispheres. While the annual-mean cyclone response to precession is
13 muted, substantial seasonal variability emerges. In the North Atlantic, the North Pacific, and the
14 Southern Ocean the response concentrates in local summer, where storm activity ranges across the
15 phases by up to 68% of its climatological mean in cyclone influence time (44% in cyclone counts),
16 while winter changes stay below about 10%. We also find summers spent near perihelion feature
17 suppressed storminess, whereas summers near aphelion experience intensified cyclone activity,
18 yielding antiphase midlatitude storm variability between the two hemispheres over the precession
19 cycle. This signal is carried by cyclone number and position, and it follows the pronounced
20 summer response of the Eady growth rate driven primarily by vertical wind shear in all three
21 basins. Monthly storm responses are phase-locked to the calendar timing of perihelion, lagged
22 by 1–2 months, consistent with surface thermal inertia. Precession therefore acts on extratropical
23 storm tracks through their quiet season. The same fingerprint emerges when the identical tracking
24 is applied to simulations of the Last Interglacial and the mid-Holocene, with weaker amplitudes
25 matching their weaker eccentricity. Summer-sensitive proxy archives should record a strong and
26 hemispherically antiphased storminess signal at the precession period that annually integrating
27 archives would miss.

28 SIGNIFICANCE STATEMENT: Long orbital cycles drive global climate variability, and the
29 strongest of these, precession, reshapes sunlight mainly in summer, while mid-latitude storms
30 happen mainly in winter. This creates an apparent disconnect between precessional forcing and
31 storm track variability. Climate model simulations covering a full precession cycle show that
32 the orbit nevertheless controls storminess by acting through summer. When perihelion falls in a
33 hemisphere's summer, that summer has fewer mid-latitude storms and the storm belt sits farther
34 poleward, and the two hemispheres evolve in antiphase through the precession cycle. Simulations
35 of the Last Interglacial, when perihelion fell in northern summer, reproduce this pattern. This
36 tells researchers where to look for orbital storm signals in geological records, namely in archives
37 sensitive to summer, not to the yearly average.

38 **1. Introduction**

39 Extratropical cyclones govern midlatitude weather systems. They transport the majority of pole-
40 ward heat and moisture outside the tropics, supply a substantial fraction of midlatitude precipitation,
41 and aggregate into the canonical storm-track pathways of the North Atlantic, North Pacific, and
42 Southern Ocean (Hoskins and Valdes 1990; Chang et al. 2002; Hoskins and Hodges 2002, 2005).
43 These cyclones develop via baroclinic instability driven by the meridional temperature gradient
44 (Eady 1949), a gradient that reaches its maximum steepness during winter. Cyclone activities
45 consequently peak in the cold season, with a consistent wintertime frequency maximum in the
46 Northern Hemisphere (Hoskins and Hodges 2002) and in the Southern Hemisphere (Simmonds
47 and Keay 2000). Midlatitude cyclogenesis is therefore most favorable during winter.

48 Precession, by contrast, exerts its primary forcing on warm seasons. Slow variations in the
49 longitude of perihelion represent the leading orbital control on seasonal insolation cycles. This
50 mechanism redistributes incoming solar radiation, and the amplitude of insolation anomalies
51 scales with a season's baseline solar irradiance (Berger 1978; Laskar et al. 2004). Midlatitude
52 summertime insolation fluctuates by over 100 W m^{-2} across a full precession cycle, whereas
53 midwinter insolation experiences only minor orbital modulation. This stark contrast explains why
54 orbital climate frameworks single out summertime solar forcing as the most impactful component of
55 climate changes (Huybers 2006). Since extratropical cyclone activity is inherently winter-favored,

56 this invites a simple expectation that precession should exert negligible control over midlatitude
57 storm tracks. But this expectation has never been put to a clean test.

58 Previous studies have looked closely at precessional signals, and they find seasonal shifts in
59 climate. This is well established in monsoon research, from the classic orbital experiments of
60 Kutzbach and Guetter (1986) to single-forcing designs that isolate precession explicitly (Bosmans
61 et al. 2015; Merlis et al. 2013), and cave records show the northern and southern monsoons
62 swinging in antiphase as perihelion migrates through the year (Wang et al. 2008). The extratropics
63 have received far less attention. Kaspar et al. (2007) simulated a stronger and poleward-shifted
64 North Atlantic winter storm track for the Eemian from two orbital time slices. Brayshaw et al.
65 (2010) linked Holocene orbital shifts to North Atlantic and Mediterranean winter storm tracks
66 via changes in baroclinicity. And Varma et al. (2012) found the Southern Hemisphere westerlies
67 migrating under Holocene orbital forcing. Explicit cyclone tracking has entered paleoclimate work
68 mainly at the Last Glacial Maximum, where ice sheets dominate the forcing (Pinto and Ludwig
69 2020; Raible et al. 2021), and in transient simulations of the last millennium (Hess and Chemke
70 2025). For tropical cyclones the precessional response has recently been isolated in high-resolution
71 paleoclimate simulations, where it is governed by atmospheric moisture (Raavi et al. 2023). Our
72 study provides the extratropical counterpart, for storms that grow by baroclinic instability instead.
73 The winter-focused precedents asked how orbital forcing moves the storm track in its strongest
74 season, and their time-slice or transient designs blend precession with obliquity, eccentricity, and
75 ice-sheet changes. To our knowledge, no previous study has applied explicit cyclone tracking
76 across a full precession cycle to isolate the orbital response of the extratropical storm tracks, and
77 none has asked which season carries that response.

78 This missing piece of research matters beyond just storm dynamics. Many paleoclimate archives
79 that record past storm activity such as wind-blown dust and sediment, or rainfall linked to westerly
80 winds only capture signals from specific seasons instead of annual averages. If a proxy record
81 mainly reflects one single season, it can either hide or falsely create an orbital signal, depending
82 on which time of year carries the strongest storm response (Laepple et al. 2011). Fully understand-
83 ing which season holds the clearest precessional storm signal is therefore essential for properly
84 interpreting these geological records. Moreover, different metrics for cyclone behavior, e.g., total
85 storm counts, how much rain each storm produces, and storm intensity, react differently to orbital

86 forcing. Researchers must evaluate these quantities separately before drawing conclusions from
87 any single geological proxy.

88 To fill this gap, we run 12 equilibrium climate simulations covering a full precession cycle.
89 Every extratropical cyclone in both hemispheres is tracked individually using 6-hour sea level
90 pressure. Our experimental design isolates precession at amplified eccentricity with all other
91 boundary conditions fixed, so that the seasonal storm response to precession forcing alone can
92 be examined. The resulting predictions are then tested against simulations of real orbital states.
93 Section 2 describes our model and simulation, and the method we use for storm tracking and
94 analysis. Section 3 presents the results, and section 4 discusses and concludes.

95 **2. Methodology**

96 *a. Model and experiments*

97 The simulations are performed with the coupled Earth system model AWI-ESM2 (Sidorenko
98 et al. 2019). The atmosphere component is ECHAM6 at T63 resolution with 47 vertical levels
99 (Stevens et al. 2013), which includes the land surface module JSBACH (Raddatz et al. 2007). The
100 ocean and sea ice component is FESOM2, a finite-volume model formulated on an unstructured
101 mesh (Danilov et al. 2017), with a spatially varying resolution of from 25 km over high latitudes
102 and coastal regions, to 150 km over open oceans.

103 12 equilibrium simulations are analyzed, taken from the idealized precession ensemble introduced
104 by Shi et al. (2025). The longitude of perihelion ω is prescribed at values from 0° to 330° in steps
105 of 30° , while eccentricity is fixed at 0.058 and obliquity at 24.5° . The eccentricity value sits
106 near the upper bound reached during the Quaternary and amplifies the precessional modulation of
107 insolation. The quoted response amplitudes are therefore upper bounds, and their scaling toward
108 lower eccentricity is not tested here. The seasonal concentration and the hemispheric antiphase
109 rest on orbital geometry alone and do not depend on this amplification. All remaining boundary
110 conditions follow the pre-industrial setup, and each phase is branched from a long pre-industrial
111 control. 30 years of 6-hourly output from the equilibrated part of each simulation form the basis
112 of the cyclone analysis. The calendar month of perihelion for each phase is computed from
113 orbital geometry (Berger 1978), and the same formulary provides the insolation fields shown in
114 Fig. 1. To test the results under realistic orbital states, we further performed three equilibrium

115 simulations with the same model, a preindustrial control, a mid-Holocene (6 ka) experiment, and
 116 a Last Interglacial (127 ka) experiment. Thirty years of six-hourly output from each are processed
 117 with the identical detection and tracking pipeline. At 127 ka perihelion falls near the June solstice
 118 with an eccentricity of 0.039, at 6 ka it falls in September, and in the preindustrial control it falls
 119 in early January.

120 *b. Cyclone detection*

121 Extratropical cyclones are detected and tracked with the algorithm of Crawford et al. (2021),
 122 and Fig. 2 summarizes the procedure with one 6-hourly field of the ensemble. 6-hourly sea level
 123 pressure is first interpolated onto 100-km equal-area grids centered on each pole, and the two
 124 hemispheres are processed separately. A grid cell qualifies as a cyclone center when its pressure
 125 p_c lies below that of all 8 neighbors and the surrounding field keeps a sufficient inward gradient.
 126 The gradient test compares p_c with the mean pressure $\bar{p}(d)$ on a one-cell ring at distance d from
 127 the candidate,

$$\frac{\bar{p}(d) - p_c}{d} \geq \gamma \quad (1)$$

128 where $d = 1000$ km and $\gamma = 750 \text{ Pa} (1000 \text{ km})^{-1}$, and candidates over terrain above 1500 m are
 129 discarded. Centers closer than 1000 km are treated as one multicenter system. The cyclone area A
 130 is obtained by growing closed isobars around each center at a 200-Pa interval until a contour fails
 131 to close, and the last closed isobar defines the edge pressure p_e (Fig. 2b). Each cyclone carries the
 132 depth and the area-equivalent radius

$$D = p_e - p_c \quad (2)$$

$$R = \sqrt{A/\pi} \quad (3)$$

133 Tracking links the centers of consecutive 6-hourly fields (Fig. 2c). For every active cyclone a
 134 first-guess position is projected from its previous displacement, damped to three quarters,

$$\mathbf{x}_q = \mathbf{x}_t + 0.75 (\mathbf{x}_t - \mathbf{x}_{t-1}) \quad (4)$$

135 and a center of the next field qualifies as the continuation when it lies within $v_{\max}\Delta t$ of both the
 136 current position $\mathbf{x}_t$ and the projection $\mathbf{x}_q$, with $v_{\max} = 150 \text{ km h}^{-1}$ and $\Delta t = 6 \text{ h}$, a search radius
 137 of 900 km per step. Among the qualifying candidates the one nearest to the projection continues
 138 the track. Centers left unmatched at the new time start tracks as genesis events, and tracks left
 139 unmatched end as lysis. Only systems poleward of 30° latitude enter the analysis.

140 The main diagnostic derived from the tracks is track density, the number of six-hourly cyclone
 141 positions falling in each grid cell per month. Tracks that merely cross the boundaries of the monthly
 142 processing windows would appear as spurious genesis or lysis events, and such truncated events
 143 (about 7% of the total) are removed. The depth, the propagation speed, and the remaining life-cycle
 144 properties of each cyclone are recorded alongside.

145 The same pipeline also yields a storm-day field, the number of days per month during which a
 146 grid point lies inside the footprint of a detected cyclone, counted from the six-hourly fields and
 147 regridded to T63. This Eulerian measure is convenient for maps of the storm distribution and of
 148 its displacement. It scales with the product of cyclone number and cyclone area, so frequency
 149 statements in this paper rest on track density.

150 All statistics rest on this single detection scheme. Absolute cyclone counts differ across tracking
 151 algorithms by a factor of two or more (Neu et al. 2013), and the strict gradient criterion of Eq. (1)
 152 keeps the census conservative. The analysis therefore reports relative responses across phases
 153 rather than absolute counts, and the tracking-free variability measure introduced below provides a
 154 check that carries no tracking assumptions.

155 An Eulerian measure of synoptic variability provides a further check that involves no tracking
 156 at all. Bandpass sea level pressure variability is computed from the 6-hourly fields with the 24-h
 157 difference filter of Wallace et al. (1988), which isolates fluctuations on the synoptic time scales that
 158 define the classical storm tracks (Blackmon 1976), and its monthly standard deviation is analyzed
 159 with the same basins, seasons, and response measures.

160 *c. Baroclinicity diagnostics*

161 The maximum Eady growth rate serves as the measure of baroclinicity (Eady 1949; Lindzen
 162 and Farrell 1980; Hoskins and Valdes 1990). It is computed as $\sigma = 0.31 |f| |\partial U / \partial z| N^{-1}$ in the
 163 925–700 hPa layer from monthly wind, temperature, and geopotential height. The response to

precession is quantified between the end members p090 and p270, which place perihelion at the June and December solstices. Because the difference fields are dipolar, plain box averages cancel, and the response amplitude is instead defined as the root-mean-square of the difference over a basin, normalized by the climatological growth rate of the season. A substitution decomposition in the style of Camargo et al. (2007) splits the difference field into a vertical-shear factor and a static-stability factor, and the share of each factor is obtained by projecting it onto the full difference pattern. The basin averages use the storm-track boxes defined in the next sub-section. This choice matters in the Southern Hemisphere, where a sector beginning at 30°S would admit the seasonal signal of the subtropical winter jet and disguise it as a storm-belt response. The shear and stability shares need not sum to one, because the two factors are not orthogonal.

d. Basins and statistics

Three storm-track basins are analyzed, the North Atlantic (30–75°N, 80°W–20°E), the North Pacific (30–75°N, 120°E–120°W), and the Southern Ocean (40–75°S, all longitudes). Basin means are area weighted. The size of the precessional response is reported as the range across the twelve phases, the maximum minus the minimum of the 30-year mean, expressed as a percentage of the phase mean. Uncertainty estimates use the interannual spread of the 30 individual years in each phase, and the significance of the end-member difference maps is assessed with a two-sided Welch t test (Welch 1947) at each grid point, treating the 30 years as independent samples; the stippling in the maps is pointwise. The local-summer contrasts between extreme phases far exceed the interannual noise, with p values below 10^{-12} in all three basins. Ranges are quoted separately for track density and for storm days, and every percentage in this paper names its metric, because the influence-time measure compounds cyclone number with cyclone area and swings harder than the count-based densities.

3. Results

a. The summer paradox

The 12 precession-phase annual midlatitude storm climatologies exhibit only minor differences, and phase anomalies relative to the multi-phase ensemble mean manifest as weak, slowly evolving dipoles aligned along major storm track pathways (Fig. 3). The basin-averaged time series quantify

192 the muted amplitude of this annual-mean signal, with across-phase ranges of annual storm days
193 of 10%, 17% and 4% in the North Atlantic, the North Pacific and the Southern Ocean, varying
194 smoothly with ω . We also observe an interhemispheric antiphase.

195 Beneath this annual background lies a pronounced seasonal cycle of storm activity. Extratropical
196 cyclone activities peak in DJF in Northern Hemisphere basins and JJA across the Southern Ocean,
197 with this seasonal timing unchanged across all orbital setups (Fig. 4a–c; Table 1). Cold-season
198 cyclones also carry greater intensity, with winter peak cyclone depths exceeding summertime
199 values by 70–100 % within the Northern Hemisphere storm basins (not shown). Overall, winter
200 dominates midlatitude cyclone activity uniformly across hemispheres and all orbital phases.

201 Precession exerts negligible control over annual-mean storm conditions but profoundly reshapes
202 seasonal cyclone variability, and its strongest modulation targets the low-activity warm season.
203 Though each precession phase retains the same wintertime peak in basin-averaged storm track
204 density, cross-phase deviations emerge almost entirely within each basin’s local summer (Fig. 4a–c).
205 Seasonal range statistics quantify this stark seasonal contrast in orbital sensitivity (Fig. 4d–f). For
206 the North Pacific, the across-phase range of track density accounts for just 7 % of the phase mean on
207 an annual basis, yet rises to 44 % in boreal summer (JJA). The North Atlantic follows an identical
208 pattern, with annual variability of 7 % versus a summertime range of 25 %. The Southern Ocean
209 mirrors this behavior but shifts the sensitive season to its local summer (DJF), with an annual
210 range of only 4 % compared to a 22 % summer signal. Cyclone system and genesis densities
211 confirm the same seasonal calendar (not shown). The influence-time diagnostic swings harder
212 still, with summertime storm-day variability reaching 68 % of the climatological mean in the North
213 Pacific (Table 1), because storm days compound cyclone number with cyclone area. All three
214 basins identify local summer as the most orbitally sensitive season in the track-based metrics, and
215 the small annual signal is what survives of a substantial summer swing after the stable winter is
216 averaged in.

217 A measure free of tracking assumptions supports this calendar with one refinement. The bandpass
218 sea level pressure variability responds to precession almost entirely in the warm half of the year
219 (Fig. A1). Over the North Pacific its summer response is nearly four times that of any other season.
220 Over the North Atlantic the peak shifts to early autumn, the calendar counterpart of the September
221 lobe of the insolation forcing, and over the Southern Ocean the storm-belt core responds most

222 strongly in local summer while the subtropical flank keeps a winter signal. Deep winter stays quiet
223 in this metric as well.

224 This summertime storm response is locked to orbital geometry. When basin-averaged track-
225 density anomalies are plotted against both calendar month and precession angle ω , the band of
226 peak anomalies traces a diagonal ridge aligned with the calendar timing of perihelion across
227 the full phase space (Fig. 4g–i), with a delay of 1–2 months. Summers spent near perihelion
228 exhibit suppressed storm activity, whereas aphelion summers feature intensified storminess. A
229 single precessional forcing thus drives opposing midlatitude storm responses between the two
230 hemispheres across the orbital cycle, as northern summer perihelion aligns with southern summer
231 aphelion under orbital precession, and vice versa, reversing the insolation control on warm-season
232 storminess between hemispheres.

233 *b. The dynamical bridge runs through summer shear*

234 The pronounced seasonal dependence of orbital storm signals originates directly from insolation
235 forcing. Averaged across the Northern Hemisphere storm-track band (40–70°N), precessional
236 insolation variability peaks at 105 W m^{-2} during boreal summer (JJA) and falls to merely 14 W m^{-2}
237 in boreal winter (DJF). The Southern Hemisphere storm band (40–70°S) exhibits mirror-symmetric
238 behavior, with a maximum seasonal forcing amplitude of 119 W m^{-2} in austral winter (DJF) versus
239 just 47 W m^{-2} in austral summer (JJA) (Fig. 1). The underlying mechanism lies in the multiplicative
240 nature of precessional insolation modulation. Shifting perihelion across the annual cycle alters
241 the Sun–Earth distance for each month, which rescales monthly incoming solar flux by a factor
242 proportional to eccentricity e (approximately $4e$, equaling 23 % for $e = 0.058$). Midwinter at
243 midlatitudes features minimal climatological insolation, leaving little baseline flux to be modulated
244 by precessional shifts. Precession forcing therefore exerts its strongest modulation on local summer
245 in each hemisphere, and barely perturbs deep winter conditions.

246 Orbital insolation anomalies are translated into extratropical storm activity via shifts in midlat-
247 itude baroclinicity. Differences in maximum Eady growth rate between the extreme precession
248 phases $\omega = 90^\circ$ and $\omega = 270^\circ$ reach their peak magnitude within each basin’s local summer (Fig. 5).
249 Normalized against seasonal climatological means, summertime Eady growth rate responses hit
250 31 % in the North Atlantic, a pronounced 92 % in the North Pacific, and 21 % in the Southern

Ocean. A decomposition analysis further reveals that vertical wind shear dominates the precessional Eady growth rate anomalies, while static stability contributes negligible signal strength. This is consistent across all storm tracks. The thermal footprint behind this response is shown in Fig. A2. Perihelion summers warm the summer hemisphere strongly, and the 850-hPa warming peaks over the subpolar continents while the ice-covered Arctic warms less. The meridional temperature gradient consequently flattens along the equatorward flank of the storm track and steepens along its poleward flank, the thermal counterpart of the growth-rate dipole, so the baroclinic source weakens in the storm-track core and shifts poleward. The Southern Hemisphere mirrors this structure in DJF with reversed sign. The storm anomalies appear somewhat poleward of the growth-rate anomalies, consistent with the downstream development of baroclinic waves (Chang et al. 2002).

c. Spatial expression, and what precession actually moves

Spatial difference maps between the two orbital end-members (p090 minus p270) yields weak responses for the annual means (Fig. 6a). By contrast, boreal summer (JJA) exhibits a prominent dipole pattern across Northern Hemisphere basins, with fewer storm days through the midlatitude core and more along the Arctic flank at perihelion summer, the signature of a poleward shift joined to an overall summer weakening. In DJF the signal migrates to the Southern Ocean and forms a circumpolar banded structure of opposite sign, marking enhanced midlatitude storm activity and an equatorward shift of the southern storm belt.

The latitudinal displacement of storm tracks driving these dipole anomalies is quantified in Fig. 7. For the North Pacific boreal summer, core storm latitude shifts from 53°N when perihelion falls in northern winter to 61°N when it falls in northern summer, yielding an 8° total poleward migration that peaks precisely at $\omega = 90^{\circ}$. The North Atlantic summertime storm track undergoes a 4.4° poleward shift following identical orbital behavior, while the Southern Ocean mirrors this response with a 3.8° poleward displacement when perihelion coincides with austral summer (near p270). A perihelion summer therefore weakens storm activity and shifts the storm population poleward, consistent with the poleward shift of baroclinic activity evident in the Eady growth rate dipole (Fig. 5a).

By contrast, the annual-mean track latitude varies by no more than 1.3° in any basin. Winter storm-track latitudes only fluctuate by $1.5\text{--}2.8^{\circ}$ across the full precession cycle, and their latitudinal

extrema do not align with solstitial orbital phases. This further confirms our earlier conclusion that midlatitude winter storms are orbitally insensitive. Spring and autumn display intermediate latitudinal variability with a coherent phase dependence. For example, autumn poleward displacement maximizes near $\omega = 150^\circ$, the orbital configuration that places perihelion in late summer. Latitudinal poleward shifts therefore track the calendar timing of perihelion identically to Fig. 4g–i, with a uniform lag of 1–2 months, consistent with surface thermal inertia. Figure 8 quantifies this basin-dependent lag. All basin-scale storm extrema cluster along basin-specific dashed lines offset from the perihelion reference. The response lags the perihelion calendar by distinct intervals across regions, near one month in the North Pacific, 1–2 months in the North Atlantic, and slightly over two months in the Southern Ocean. The 30° phase spacing resolves these lags to about one month.

d. What the rain records

The hydrological signature of precessional forcing evolves in antiphase with storm dynamical responses (Fig. 9). Detected extratropical cyclones contribute roughly one-third of total midlatitude precipitation across our simulations, consistent with storm precipitation partitioning derived from observational reanalysis datasets (Hawcroft et al. 2012). During boreal summer over the North Pacific, mean precipitation per individual storm rises by as much as 15 % at the orbital phase where total storm frequency reaches its minimum, yielding nearly perfectly opposing variability between the two metrics. Perihelion summers feature warmer, more humid midlatitude atmospheres, which amplifies precipitation intensity within the remaining cyclones; nevertheless, the sharp reduction in storm number dominates the total signal, such that basin-wide summertime precipitation drops by 17 % between the two extreme orbital phases. The fractional share of precipitation originating from cyclones also declines markedly, with a full-cycle range of 9.5 % in North Pacific summer compared to just 3 % for the annual mean. Precessional forcing thus exerts two counteracting controls on extratropical hydroclimate simultaneously. It reduces total cyclone counts while boosting precipitation intensity per storm.

4. Discussion

The seasonal preference seen in storm responses originates purely from orbital geometry, independent of model physics. Precession redistributes insolation in proportion to the insolation a

season already has, meaning winter hemispheres have very little background insolation for orbital changes to modify (Berger 1978). Fig. 1 clearly illustrates this stark seasonal contrast in forcing strength. Winter insolation varies by only roughly 20 W m^{-2} across the precession cycle, while summer values exceed 125 W m^{-2} within mid-latitude storm-track latitudes. This is model independent and will hold in any climate models as it simply reflects the orbital geometry behind the well-known summer-dominated orbital forcing framework (Huybers 2006). Existing monsoon studies have long confirmed that summer bears the strongest precessional signal (Kutzbach and Guetter 1986; Bosmans et al. 2015). But the new insight of our work lies elsewhere. This summer-focused orbital forcing acts on a winter-dominated weather system, reshaping storm activity exclusively during its normally quiet warm season. Note that the term “quiet” here refers to the weaker overall intensity of the summer storm track, not its population. Both hemispheres still host abundant storm systems in local summer, and they are merely weaker, shallower, and slower moving than winter cyclones.

The mechanism that converts summer insolation into summer storms is the thermal wind. When perihelion falls in local summer, the corresponding hemisphere warms, most strongly over the subpolar continents. The meridional temperature gradient flattens along the equatorward flank of the storm belt, and the vertical wind shear that drives baroclinic growth weakens in the storm-track core (Fig. A2). Our factor decomposition analysis confirms that the shear term carries essentially the whole Eady growth rate response in all three ocean basins, while static stability plays a negligible role. We see similar physical mechanism operating in the modern Northern Hemisphere, where greenhouse warming takes the place of perihelion-driven heating. Stronger summer warming at high latitudes has weakened summertime large-scale circulation, eddy kinetic energy, and the number of strong summer cyclones (Coumou et al. 2015; Chang et al. 2016; Gertler and O’Gorman 2019). Future climate projections show the same shear-based decline of Northern Hemisphere summer cyclone activity and baroclinic energy as the globe warms (Lehmann et al. 2014; Priestley and Catto 2022). Two differences should be noted. Greenhouse warming is a transient forcing that both hemispheres feel in phase, whereas our simulations are equilibrated and precession swings the hemispheres in antiphase, which makes precession the cleaner natural experiment for the summer shear mechanism. Even so, both forcings follow one unified rule that is the core theme of this study: cooler summers host more storm activity.

338 Compared to summer, storm systems in winter are much more stable. Across all three ocean
339 basins, winter storm activity changes by less than 10% over a full precession cycle, even though
340 winter Eady growth rates shift by 16–18% between the two orbital end-members, and weak winter
341 insolation forcing cannot fully explain this stability. The winter storm track appears saturated, in
342 the sense that ample baroclinicity is available in every phase and moderate changes in the growth
343 rate do not translate into changes in occurrence, though a winter occurrence that simply responds
344 less sensitively to growth rate would produce the same numbers and cannot be excluded with
345 twelve phases. A transient simulation of the last millennium likewise found cyclone counts nearly
346 insensitive to external forcing (Raible et al. 2018; Hess and Chemke 2025). Our findings expand
347 this conclusion to the far larger swings in solar radiation driven by precession, and this insensitivity
348 only holds for wintertime conditions.

349 Geological records sensitive to summertime conditions within midlatitude storm belts should
350 preserve prominent precession-scale storm signals, whereas proxies that average climate signals
351 across winter or the full year will show only minor orbital imprint. Just as observed for temperature
352 reconstructions, the seasonal bias of a given archive can either generate artificial orbital storm
353 signals or fully obscure true precessional variations (Laepple et al. 2011). Furthermore, summer-
354 sensitive storm records from both hemispheres will fluctuate oppositely over precessional cycles,
355 forming an extratropical equivalent of the well-documented monsoon seesaw (Wang et al. 2008;
356 Erb et al. 2015). These statements combine into a three-part test. A genuine precessional storm
357 signal should appear at the precession period, it should be antiphased between the hemispheres,
358 and it should stand out in summer-biased archives while staying weak in winter-biased archives of
359 the same region. Precipitation-based archives blend the count signal with the opposing per-storm
360 rainfall response described below, so the cleanest targets are proxies that sense storm occurrence or
361 storm-belt position rather than rain amount. Current published sediment records already support
362 the feasibility of testing this framework. Distinct precession-period signals have been detected in
363 Northern Hemisphere midlatitude westerly proxies (Li et al. 2019), as well as North Pacific dust
364 records. Precessional variation appears specifically in moisture-sensitive dust fractions, while the
365 annually averaged dust flux responds primarily to obliquity forcing (Zhong et al. 2024). Meanwhile,
366 dust intensity archives that capture winter-only signals display very weak precessional power (Chen
367 et al. 2014). Regarding the Southern Hemisphere, subtropical margins of the southern westerly belt

368 exhibit clear precession cycles (Lamy et al. 2019), consistent with our results. And the presence
369 or absence of precession signals at individual sites depends on their latitudinal position within the
370 wind belt (Kaiser et al. 2024). Records of the southern westerlies also show orbital-band migrations
371 of the belt itself (Lamy et al. 2010; Varma et al. 2012), the positional counterpart of the poleward
372 summer shift in Fig. 7, and glacial-state cyclone modeling has shown how strongly reconstructed
373 storminess depends on which cyclone property a proxy senses (Pinto and Ludwig 2020).

374 The prediction can already be tested against the simulations of real orbital states introduced in
375 section 2, a preindustrial control, a mid-Holocene experiment, and a Last Interglacial experiment
376 performed with the same model and tracked with the same pipeline (Figs. 10 and 11). At 127 ka
377 perihelion falls near the June solstice, and the tracked response reproduces our template. Summer
378 track density drops by 8% in the North Atlantic and 11% in the North Pacific relative to the
379 preindustrial run, winter changes stay within about 3%, austral summer track density over the
380 Southern Ocean rises by 8%, and the northern summer storm tracks shift poleward by one to two
381 degrees while the austral summer belt shifts equatorward (Fig. 10). The summer difference maps
382 show the same weakening and poleward displacement of the northern storm tracks that the amplified
383 end members produce (Fig. 11). At 6 ka the reading is different but equally instructive. Perihelion
384 falls in September, both solstice levers are nearly neutral, and the calendar forcing reduces to a
385 warm September paired with a cool March. The mid-Holocene response follows exactly that pair,
386 with fewer and poleward-shifted storms in autumn and the reverse tendency in spring, the shoulder
387 seasons that the forcing analysis of section 3 anticipates. The response amplitudes are roughly
388 half of our amplified end-member contrast, in line with the weaker eccentricity of the real orbits.
389 These experiments combine realistic precession with their own obliquity and greenhouse settings,
390 so they test the prediction under realistic combined forcing rather than isolating precession.

391 Although the three storm basins follow the same physical mechanism, they exhibit different
392 regional behaviors, which can be distinguished through separate diagnostic metrics. In local sum-
393 mer, storm frequency and track position vary substantially, by 16–68% across the precession cycle
394 depending on basin and metric, while storm propagation speed remains largely unchanged. By
395 contrast, storm intensity exhibits clear basin-dependent differences. The Southern Ocean shows a
396 22% swing in storm frequency but only a 4% change in storm intensity. The North Pacific features
397 a systematic reduction in overall storm activity during perihelion summers. This is consistent with

398 previous findings from anthropogenic warming scenarios that storm frequency and intensity re-
399 spond to external forcing through independent physical mechanism (Bengtsson et al. 2009; Ulbrich
400 et al. 2009; Catto et al. 2019). Hydrological responses further amplify this contrast. Individual
401 storms produce stronger precipitation in summers with reduced cyclone numbers, indicating that
402 hydrological adjustments exceed dynamical changes under orbital forcing, a pattern analogous to
403 that identified under greenhouse warming (Bengtsson et al. 2009). Given that extratropical precip-
404 itation is predominantly generated by cyclones and frontal systems (Hawcroft et al. 2012; Catto and
405 Pfahl 2013), this competing effect substantially alters precipitation partitioning. A precipitation
406 proxy therefore tends to record the combined signal of reduced storm occurrence and intensified
407 single-storm rainfall during perihelion summers.

408 Our study is based on idealized simulations designed to isolate pure precessional forcing. The
409 experimental setup holds eccentricity, obliquity, ice sheets, and greenhouse gas concentrations fixed
410 and does not incorporate transient climate evolution or orbital cross-interactions. As such, our
411 results represent dynamical responses to idealized orbital conditions rather than reconstructions
412 of specific geological epochs. Three further limitations deserve note. The T63 atmosphere
413 underestimates absolute cyclone depth and count, though the across-phase contrasts that carry
414 our conclusions are less sensitive to this bias. All statistics rest on one tracking scheme, with
415 the tracking-free variability check as the main guard. And the amplified eccentricity widens the
416 lever, so the quoted amplitudes are upper bounds. The realistic-orbit comparison of Fig. 10, with
417 roughly half the lever and roughly half the response, offers a first indication that the response
418 scales with the forcing, though a dedicated eccentricity ladder is still missing. Isolating the direct
419 radiative response from coupled ocean and sea-ice feedbacks with atmosphere-only experiments
420 is a natural next step. Future work could incorporate time-varying boundary conditions and full
421 orbital combinations to further generalize the storm seasonal response across broader paleoclimate
422 backgrounds.

423 5. Conclusions

424 In our study, we carry out a set of equilibrium climate simulations covering a full precession
425 cycle, with explicit tracking of all extratropical cyclones across both hemispheres to isolate pre-
426 cessional forcing. We find that precession regulates extratropical cyclones primarily through local

427 summer, the season with naturally low storm activity, while winter storm tracks show only minor
428 changes across all precession phases. Storm activity calms down when perihelion falls in local
429 summer and intensifies when aphelion coincides with local summer. As perihelion summer in one
430 hemisphere matches aphelion summer in the other, mid-latitude storminess of the two hemispheres
431 varies in antiphase during a precession cycle. This orbital signal propagates through shear-driven
432 baroclinicity and expresses itself in cyclone numbers and position, both of which are phase locked
433 to the calendar month of perihelion, with a delay of 1–2 months. Our results also bring clear
434 implications for paleoclimate proxy interpretation. Orbital storm signals should be extracted from
435 midlatitude archives sensitive to summer climate, and the recorded signals will display opposite
436 variations between the two hemispheres. Applying the same tracking to simulations of the mid-
437 Holocene and the Last Interglacial reproduces the predicted pattern at reduced amplitude, in line
438 with their weaker orbital lever.

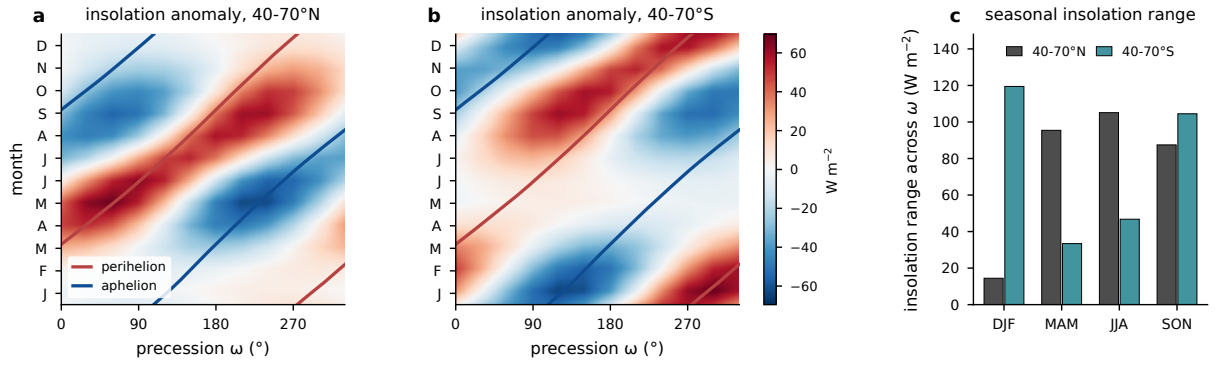


FIG. 1. Precessional insolation forcing from orbital geometry with eccentricity 0.058 and obliquity 24.5° .
 (a) Monthly insolation anomaly relative to the phase mean, averaged over the storm-track band $40\text{--}70^\circ\text{N}$, as
 a function of calendar month and precession phase ω , with the perihelion (red) and aphelion (blue) calendar
 months of each phase overlaid. (b) As in (a), but for $40\text{--}70^\circ\text{S}$. (c) Seasonal forcing amplitude in the two bands,
 the seasonal mean of the monthly maximum minus minimum insolation across the twelve phases. Units: W m^{-2} .

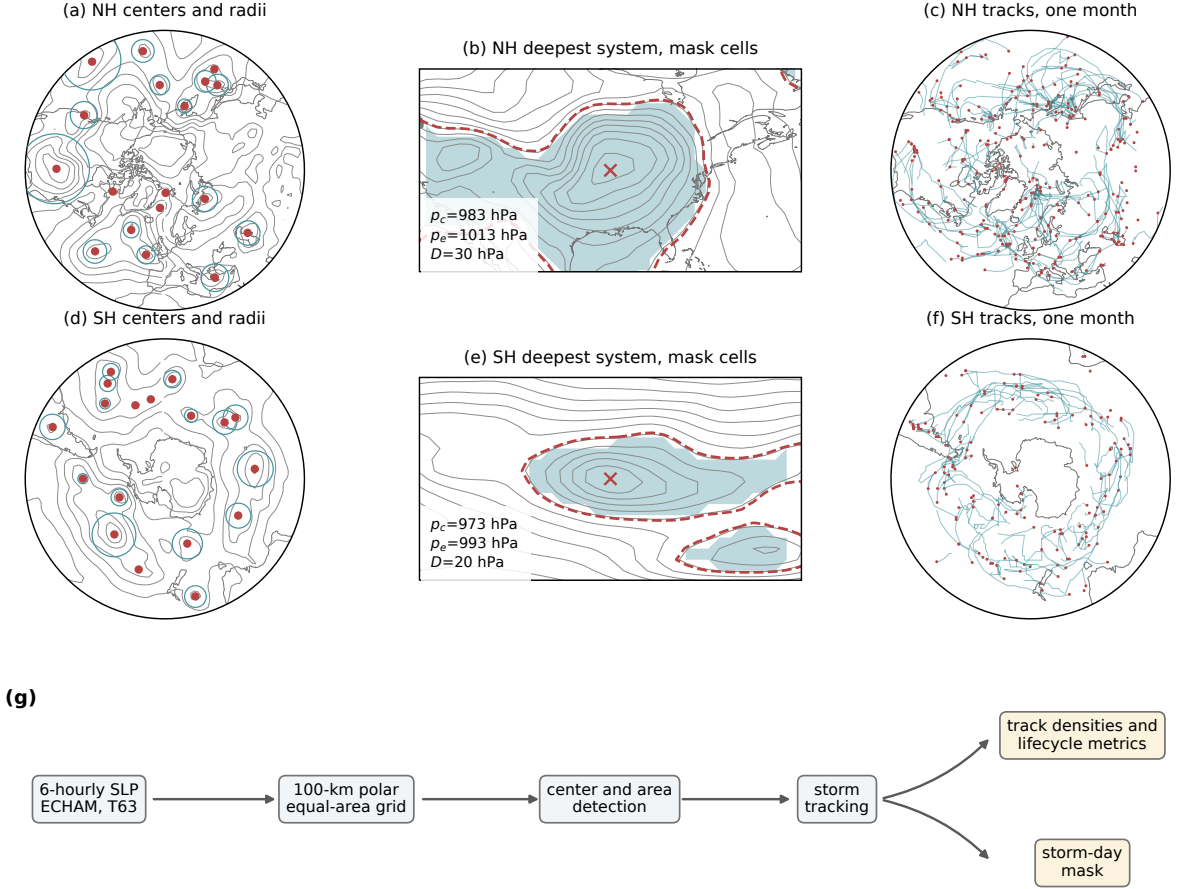


FIG. 2. Cyclone detection and tracking procedure, illustrated with one six-hourly field of phase p000 (15 January of one model year). (a) Sea level pressure (contours, 8-hPa interval) with all detected Northern Hemisphere cyclone centers poleward of 30°N (dots) and their area-equivalent radii $R = \sqrt{A/\pi}$ (circles). (b) The deepest Northern Hemisphere system of that field, with the outermost closed isobar p_e (dashed), the cyclone center (cross), the central and edge pressures and depth annotated, and the binary mask cells (shading), defined by $SLP < p_e$ inside the $\sqrt{A}$ search window; contour interval 4 hPa. (c) All Northern Hemisphere cyclone tracks of the month (lines) with genesis points (dots). (d)–(f) As in (a)–(c), but for the Southern Hemisphere. (g) Processing chain.

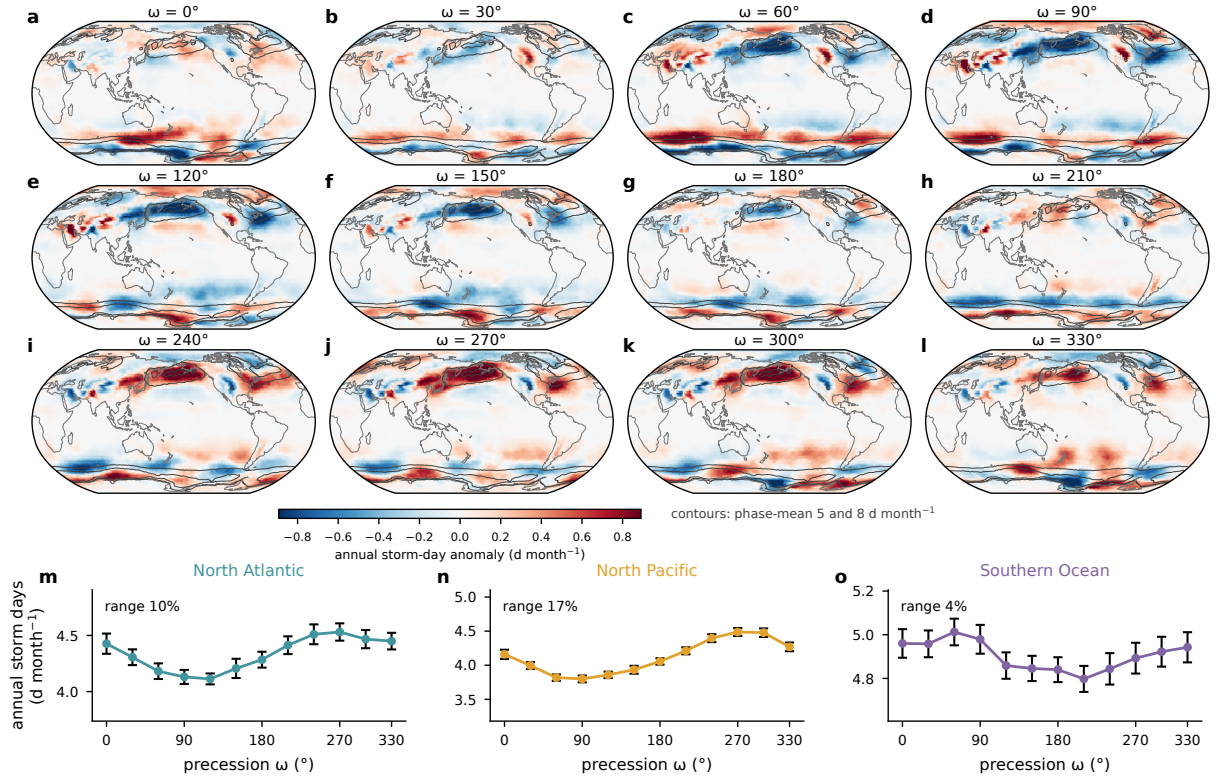


FIG. 3. Annual-mean storm days across the precession cycle. (a)–(l) Annual storm-day anomaly of each phase relative to the twelve-phase mean (shading), with the phase-mean climatology outlined at 5 and 8 d month⁻¹ (contours). (m)–(o) Basin-mean annual storm days against ω with interannual standard errors; the across-phase range as a percentage of the phase mean is annotated in each panel.

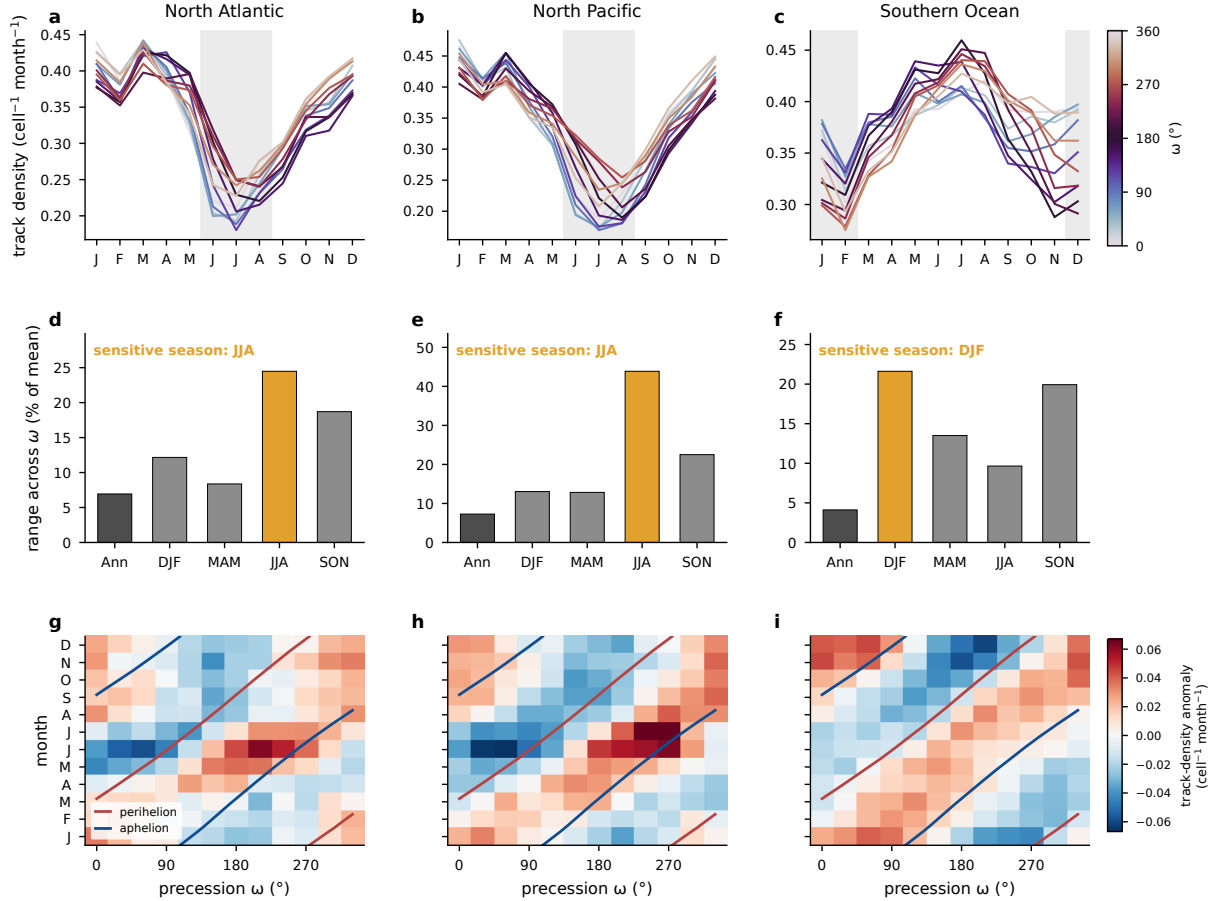


FIG. 4. Seasonal concentration of the precessional storm response, from the Lagrangian tracks. (a)–(c) Basin-mean track density (cell⁻¹ month⁻¹) by calendar month for the North Atlantic, the North Pacific, and the Southern Ocean, one line per phase colored by ω , with local summer shaded. (d)–(f) Across-phase range of track density (maximum minus minimum, percent of the phase mean) for the annual mean and the four seasons, with the local-summer bar highlighted. (g)–(i) Basin-mean track-density anomaly relative to the phase mean as a function of calendar month and precession phase (unsmoothed), with the perihelion (red) and aphelion (blue) calendar months of each phase overlaid.

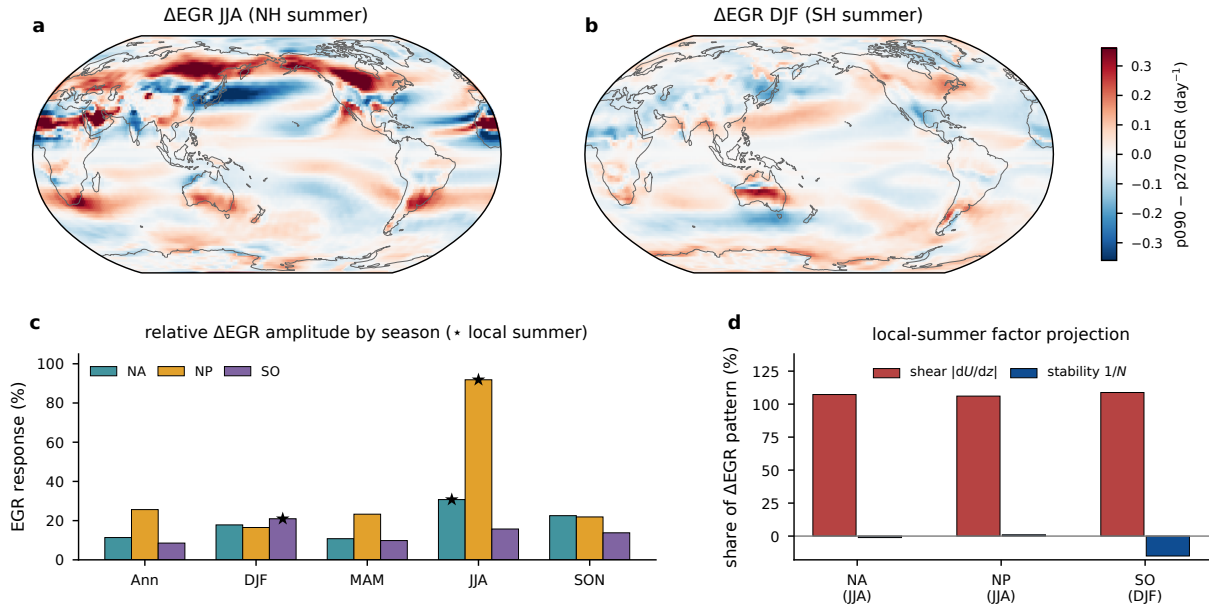


FIG. 5. Precessional response of the maximum Eady growth rate (925–700 hPa). (a) p090 minus p270 difference for JJA and (b) for DJF (day^{-1}). (c) Seasonal response amplitude in each basin, the RMS of the p090 minus p270 difference over the storm box as a percentage of the seasonal climatology, with stars marking local summer. (d) Projection shares of the vertical-shear and static-stability factors in the local-summer difference pattern.

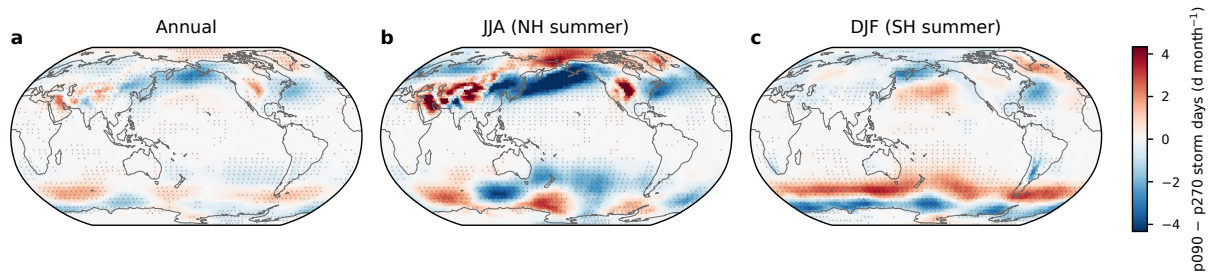


FIG. 6. Difference in storm days (d month^{-1}) between p090 and p270 for (a) the annual mean, (b) JJA, and (c) DJF, on a common color scale. Stippling marks grid points where the difference is significant at the 95% level based on a two-sided Welch t test over the 30 simulated years.

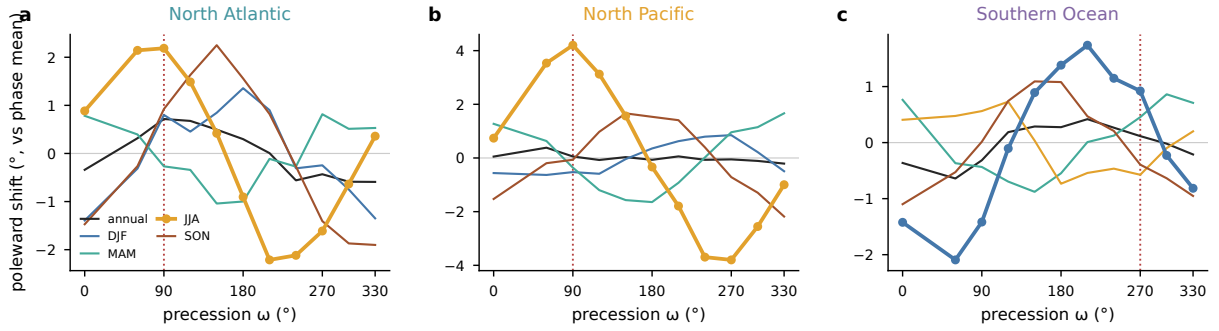


FIG. 7. Storm-track latitude across the precession cycle, the storm-day-weighted mean latitude over each basin sector shown as the poleward anomaly from each season's phase mean, for (a) the North Atlantic, (b) the North Pacific, and (c) the Southern Ocean. Lines give the annual mean and the four seasons, with the local summer of each basin drawn heavy with markers. Vertical dotted lines mark the phase with perihelion in the local summer of each basin.

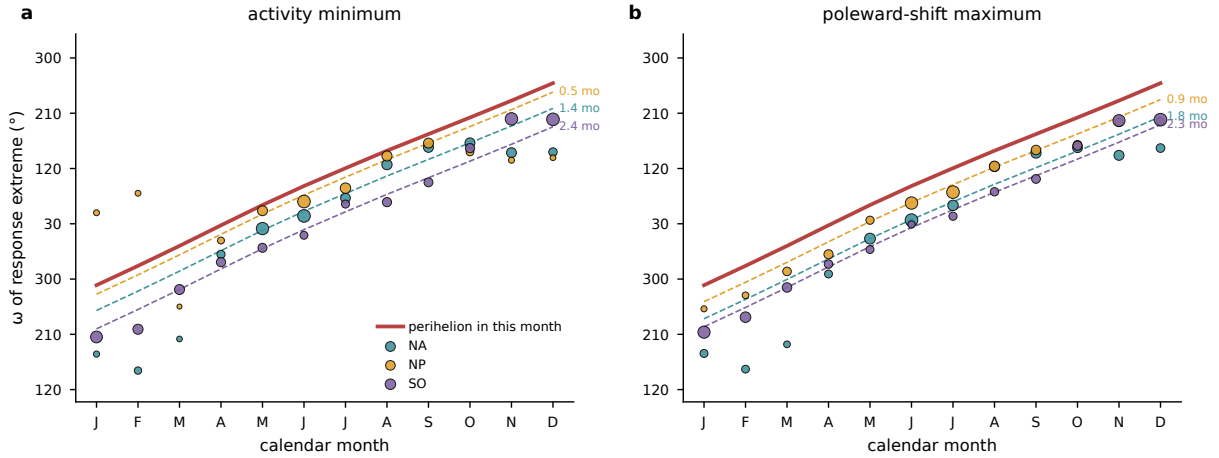


FIG. 8. Phase locking of the storm response to the perihelion calendar. For each calendar month and basin, the phase of the first harmonic across the 12 precession phases of (a) basin-mean storm days, shown as the ω of the activity minimum, and (b) the storm-day-weighted track latitude, shown as the ω of the poleward maximum. Marker size scales with the harmonic amplitude. The red line gives the ω that places perihelion in that calendar month, and dashed lines repeat it delayed by each basin's median lag over the amplitude-strong months (annotated in months).

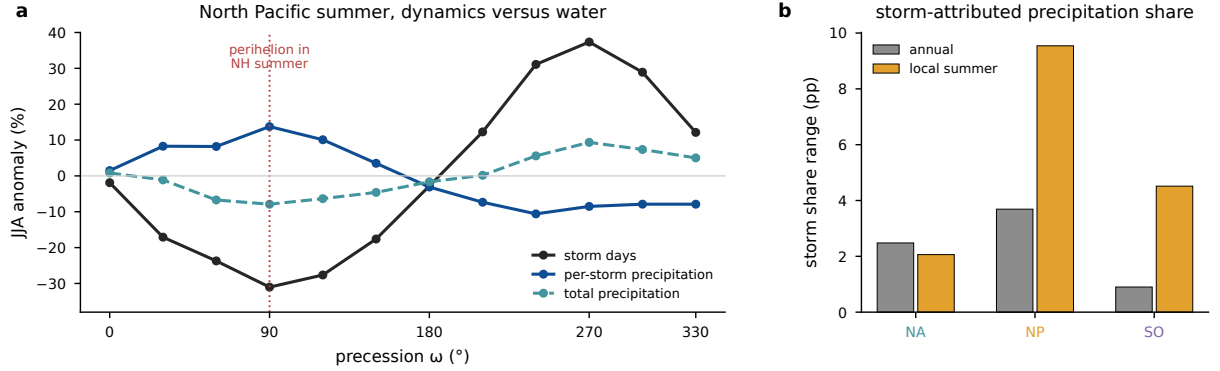


FIG. 9. Storm-attributed precipitation across the precession cycle. (a) North Pacific JJA percent anomalies from the phase mean of basin-mean storm days, per-storm precipitation, and total precipitation; the dotted line marks the phase with perihelion in NH summer. (b) Across-phase range of the storm-attributed share of precipitation, accumulated precipitation inside storm footprints divided by total accumulated precipitation (percentage points), for the annual mean and local summer in each basin.

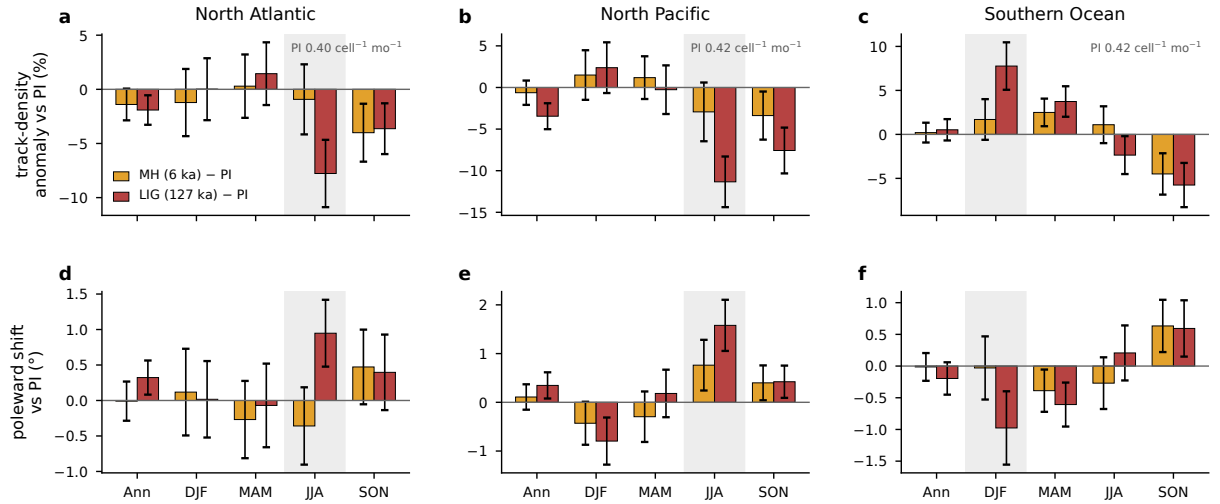
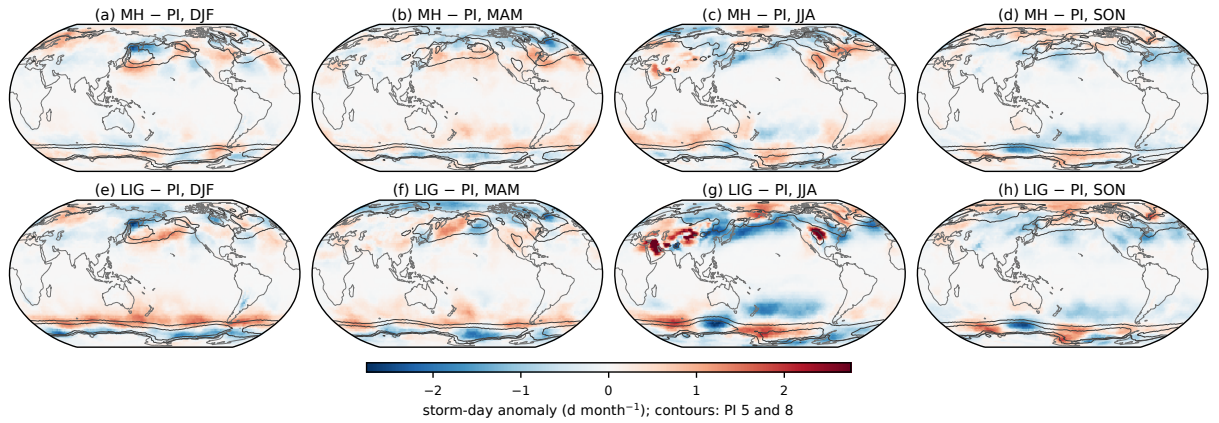


FIG. 10. Seasonal storm response of the real-orbit experiments, tracked with the same pipeline as the main ensemble and referenced to the preindustrial control. (a)–(c) Anomalies of basin-mean track density for the mid-Holocene (gold) and the Last Interglacial (red), as percentages of the preindustrial mean, for the annual mean and the four seasons, with error bars giving twice the standard error of the 30-year difference and the local summer shaded. Track density is normalized per T63 grid cell, and the annotated preindustrial values give the absolute scale in the units of Fig. 4. (d)–(f) The corresponding change in the track-point mean latitude, positive poleward.



553 FIG. 11. Storm-day anomalies of the real-orbit experiments relative to the preindustrial control, from the
 554 same storm-day product as the main ensemble. (a)–(d) Mid-Holocene minus preindustrial for DJF, MAM, JJA,
 555 and SON. (e)–(h) Last Interglacial minus preindustrial. Contours outline the preindustrial climatology at 5 and
 556 8 d month^{−1}. The isolated warm-season maxima over subtropical continental interiors reflect stationary surface
 557 heat lows rather than traveling cyclones.

563 TABLE 1. Three-basin comparison of the precessional storm response. Ranges are max minus min across
564 the twelve phases as a percentage of the phase mean. Storm and sensitive seasons refer to local winter and
565 local summer. The storm season is diagnosed from cyclogenesis counts, the sensitive season from the seasonal
566 storm-day range. The EGR response is the RMS p090–p270 difference over the basin sector relative to its
567 climatology, and the shear share is the projection of the vertical-shear factor onto the response pattern.

Basin	Storm season	Sensitive season	Storm-day range ann/summer (%)	Summer max/min ω (°)	EGR response summer (%)	Shear share (%)
North Atlantic	DJF	JJA	10 / 32	240 / 90	31	107
North Pacific	DJF	JJA	17 / 68	270 / 90	92	106
Southern Ocean	JJA	DJF	4 / 16	60 / 210	21	109

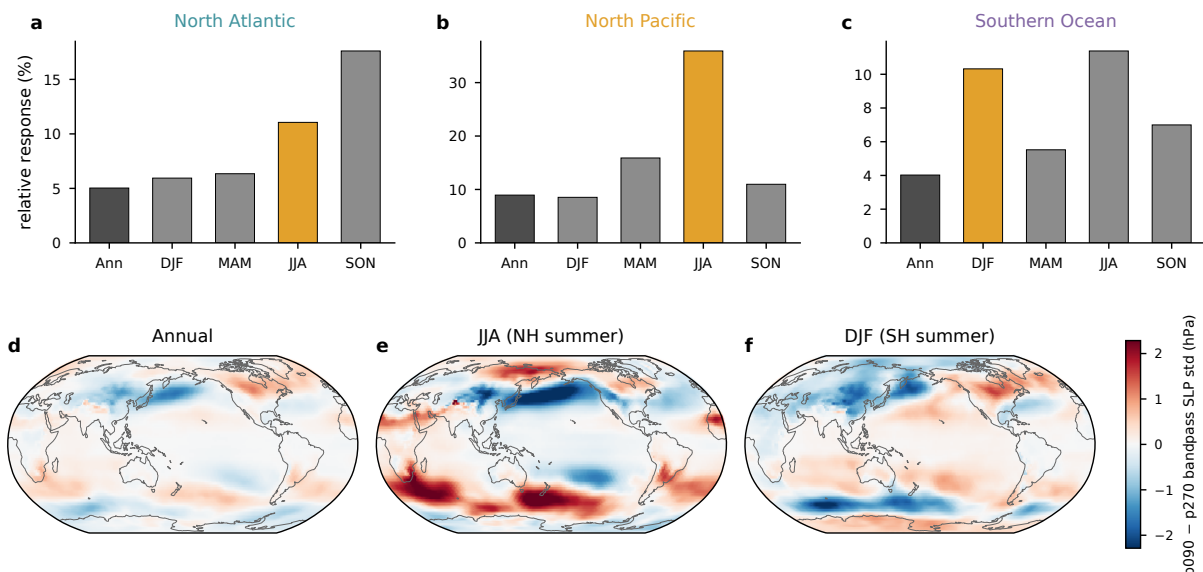


FIG. A1. Tracking-free Eulerian storm-track metric, the monthly standard deviation of 24-h differences of six-hourly sea level pressure. (a)–(c) Relative seasonal response in each basin, the RMS of the p090 minus p270 difference over the storm box as a percentage of the seasonal climatology (as in Fig. 5c), with the local-summer bar highlighted. (d)–(f) The p090 minus p270 difference of the same quantity for the annual mean, JJA, and DJF (hPa).

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Data availability statement. The AWI-ESM simulations of the idealized precession ensemble are described in Shi et al. (2025). The preindustrial, mid-Holocene, and Last Interglacial simulations are available through the Earth System Grid Federation. The cyclone-track database, the gridded storm statistics, and the analysis scripts underlying all figures will be archived and assigned a DOI upon acceptance. [TODO repository DOI.]

APPENDIX

Tracking-independent and thermal diagnostics

Figures A1 and A2 present the tracking-free bandpass diagnostic and the thermal footprint of the orbital end members discussed in sections 2 and 3.

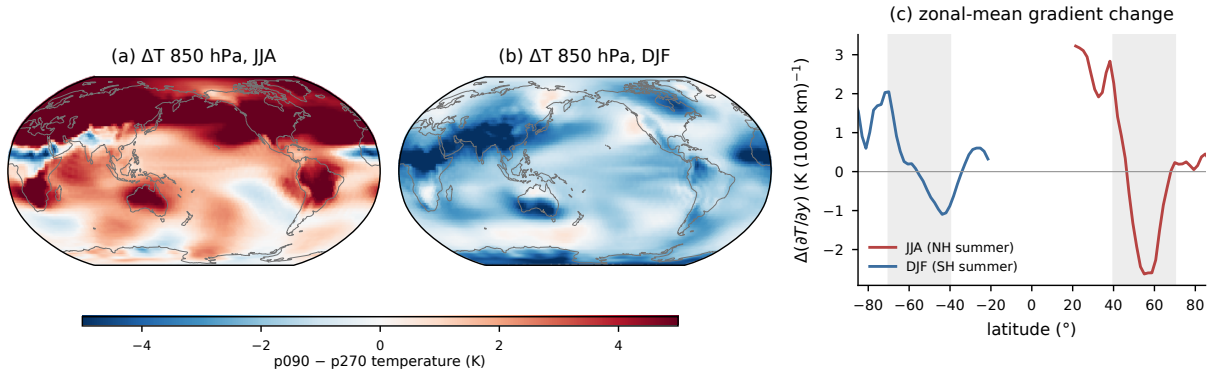


FIG. A2. Thermal footprint of the orbital end members. (a) p090 minus p270 temperature at 850 hPa for JJA and (b) for DJF (K). (c) Change in the zonal-mean meridional temperature gradient for JJA (red) and DJF (blue), with the 40–70° storm bands shaded. In the Northern Hemisphere negative values mark a steepening of the poleward temperature decline, and in the Southern Hemisphere positive values do.

References

- Bengtsson, L., K. I. Hodges, and N. Keenlyside, 2009: Will extratropical storms intensify in a warmer climate? *Journal of Climate*, **22** (9), 2276–2301, <https://doi.org/10.1175/2008JCLI2678.1>.
- Berger, A. L., 1978: Long-term variations of daily insolation and quaternary climatic changes. *Journal of the Atmospheric Sciences*, **35** (12), 2362–2367, [https://doi.org/10.1175/1520-0469\(1978\)035\(2362:LTVODI\)2.0.CO;2](https://doi.org/10.1175/1520-0469(1978)035(2362:LTVODI)2.0.CO;2).
- Blackmon, M. L., 1976: A climatological spectral study of the 500 mb geopotential height of the Northern Hemisphere. *Journal of the Atmospheric Sciences*, **33** (8), 1607–1623, [https://doi.org/10.1175/1520-0469\(1976\)033\(1607:ACSSOT\)2.0.CO;2](https://doi.org/10.1175/1520-0469(1976)033(1607:ACSSOT)2.0.CO;2).
- Bosmans, J. H. C., S. S. Drijfhout, E. Tuenter, F. J. Hilgen, L. J. Lourens, and E. J. Rohling, 2015: Response of the north african summer monsoon to precession and obliquity forcings in the earth gcm. *Climate Dynamics*, **44** (1–2), 279–297, <https://doi.org/10.1007/s00382-014-2260-z>.
- Brayshaw, D. J., B. Hoskins, and E. Black, 2010: Some physical drivers of changes in the winter storm tracks over the North Atlantic and Mediterranean during the Holocene. *Philosophical Transactions of the Royal Society A*, **368** (1931), 5185–5223, <https://doi.org/10.1098/rsta.2010.0180>.
- Camargo, S. J., K. A. Emanuel, and A. H. Sobel, 2007: Use of a genesis potential index to diagnose ENSO effects on tropical cyclone genesis. *Journal of Climate*, **20** (19), 4819–4834, <https://doi.org/10.1175/JCLI4282.1>.
- Catto, J. L., and S. Pfahl, 2013: The importance of fronts for extreme precipitation. *Journal of Geophysical Research: Atmospheres*, **118** (19), 10 791–10 801, <https://doi.org/10.1002/jgrd.50852>.
- Catto, J. L., and Coauthors, 2019: The future of midlatitude cyclones. *Current Climate Change Reports*, **5**, 407–420, <https://doi.org/10.1007/s40641-019-00149-4>.
- Chang, E. K. M., S. Lee, and K. L. Swanson, 2002: Storm track dynamics. *Journal of Climate*, **15** (16), 2163–2183, [https://doi.org/10.1175/1520-0442\(2002\)015\(02163:STD\)2.0.CO;2](https://doi.org/10.1175/1520-0442(2002)015(02163:STD)2.0.CO;2).

- Chang, E. K. M., C.-T. Ma, C. Zheng, and A. M. W. Yau, 2016: Observed and projected decrease in Northern Hemisphere extratropical cyclone activity in summer and its impacts on maximum temperature. *Geophysical Research Letters*, **43** (5), 2200–2208, <https://doi.org/10.1002/2016GL068172>.
- Chen, J., X. Liu, and V. A. Kravchinsky, 2014: Response of the high-resolution Chinese loess grain size record to the 50°N integrated winter insolation during the last 500,000 years. *Geophysical Research Letters*, **41** (17), 6244–6251, <https://doi.org/10.1002/2014GL060239>.
- Coumou, D., J. Lehmann, and J. Beckmann, 2015: The weakening summer circulation in the Northern Hemisphere mid-latitudes. *Science*, **348** (6232), 324–327, <https://doi.org/10.1126/science.1261768>.
- Crawford, A. D., E. A. P. Schreiber, N. Sommer, M. C. Serreze, J. C. Stroeve, and D. G. Barber, 2021: Sensitivity of Northern Hemisphere cyclone detection and tracking results to fine spatial and temporal resolution using ERA5. *Monthly Weather Review*, **149** (8), 2581–2598, <https://doi.org/10.1175/MWR-D-20-0417.1>.
- Danilov, S., D. Sidorenko, Q. Wang, and T. Jung, 2017: The Finite-volume Sea ice–Ocean Model (FESOM2). *Geoscientific Model Development*, **10** (2), 765–789, <https://doi.org/10.5194/gmd-10-765-2017>.
- Eady, E. T., 1949: Long waves and cyclone waves. *Tellus*, **1** (3), 33–52, <https://doi.org/10.3402/tellusa.v1i3.8507>.
- Erb, M. P., C. S. Jackson, and A. J. Broccoli, 2015: Using single-forcing gcm simulations to reconstruct and interpret quaternary climate change. *Journal of Climate*, **28** (24), 9746–9767, <https://doi.org/10.1175/JCLI-D-15-0329.1>.
- Gertler, C. G., and P. A. O’Gorman, 2019: Changing available energy for extratropical cyclones and associated convection in Northern Hemisphere summer. *Proceedings of the National Academy of Sciences*, **116** (10), 4105–4110, <https://doi.org/10.1073/pnas.1812312116>.
- Hawcroft, M. K., L. C. Shaffrey, K. I. Hodges, and H. F. Dacre, 2012: How much Northern Hemisphere precipitation is associated with extratropical cyclones? *Geophysical Research Letters*, **39**, L24 809, <https://doi.org/10.1029/2012GL053866>.

- Hess, O., and R. Chemke, 2025: Forced changes in the North Atlantic storm track over the last millennium. *Geophysical Research Letters*, **52**, e2025GL115 894, <https://doi.org/10.1029/2025GL115894>.
- Hoskins, B. J., and K. I. Hodges, 2002: New perspectives on the Northern Hemisphere winter storm tracks. *Journal of the Atmospheric Sciences*, **59** (6), 1041–1061, [https://doi.org/10.1175/1520-0469\(2002\)059<1041:NPOTNH>2.0.CO;2](https://doi.org/10.1175/1520-0469(2002)059<1041:NPOTNH>2.0.CO;2).
- Hoskins, B. J., and K. I. Hodges, 2005: A new perspective on Southern Hemisphere storm tracks. *Journal of Climate*, **18** (20), 4108–4129, <https://doi.org/10.1175/JCLI3570.1>.
- Hoskins, B. J., and P. J. Valdes, 1990: On the existence of storm-tracks. *Journal of the Atmospheric Sciences*, **47** (15), 1854–1864, [https://doi.org/10.1175/1520-0469\(1990\)047<1854:OTEOST>2.0.CO;2](https://doi.org/10.1175/1520-0469(1990)047<1854:OTEOST>2.0.CO;2).
- Huybers, P., 2006: Early pleistocene glacial cycles and the integrated summer insolation forcing. *Science*, **313** (5786), 508–511, <https://doi.org/10.1126/science.1125249>.
- Kaiser, J., E. Schefuß, J. A. Collins, and Coauthors, 2024: Orbital modulation of subtropical versus subantarctic moisture sources in the southeast Pacific mid-latitudes. *Nature Communications*, **15**, 7512, <https://doi.org/10.1038/s41467-024-51985-4>.
- Kaspar, F., T. Spanghel, and U. Cubasch, 2007: Northern hemisphere winter storm tracks of the eemian interglacial and the last glacial inception. *Climate of the Past*, **3** (2), 181–192, <https://doi.org/10.5194/cp-3-181-2007>.
- Kutzbach, J. E., and P. J. Guetter, 1986: The influence of changing orbital parameters and surface boundary conditions on climate simulations for the past 18,000 years. *Journal of the Atmospheric Sciences*, **43** (16), 1726–1759, [https://doi.org/10.1175/1520-0469\(1986\)043<1726:TIOCOP>2.0.CO;2](https://doi.org/10.1175/1520-0469(1986)043<1726:TIOCOP>2.0.CO;2).
- Laepfle, T., M. Werner, and G. Lohmann, 2011: Synchronicity of antarctic temperatures and local solar insolation on orbital timescales. *Nature*, **471** (7336), 91–94, <https://doi.org/10.1038/nature09825>.

- 678 Lamy, F., R. Kilian, H. W. Arz, J.-P. Francois, J. Kaiser, M. Prange, and T. Steinke, 2010: Holocene
679 changes in the position and intensity of the southern westerly wind belt. *Nature Geoscience*,
680 **3 (10)**, 695–699, <https://doi.org/10.1038/ngeo959>.
- 681 Lamy, F., and Coauthors, 2019: Precession modulation of the South Pacific westerly wind belt
682 over the past million years. *Proceedings of the National Academy of Sciences*, **116 (47)**, 23 455–
683 23 460, <https://doi.org/10.1073/pnas.1905847116>.
- 684 Laskar, J., P. Robutel, F. Joutel, M. Gastineau, A. C. M. Correia, and B. Levrard, 2004: A long-
685 term numerical solution for the insolation quantities of the earth. *Astronomy and Astrophysics*,
686 **428 (1)**, 261–285, <https://doi.org/10.1051/0004-6361:20041335>.
- 687 Lehmann, J., D. Coumou, K. Frieler, A. V. Eliseev, and A. Levermann, 2014: Future changes
688 in extratropical storm tracks and baroclinicity under climate change. *Environmental Research*
689 *Letters*, **9 (8)**, 084 002, <https://doi.org/10.1088/1748-9326/9/8/084002>.
- 690 Li, Y., Y. Song, Q. Yin, L. Han, and Coauthors, 2019: Orbital and millennial northern mid-latitude
691 westerlies over the last glacial period. *Climate Dynamics*, **53 (5–6)**, 3315–3324, [https://doi.org/](https://doi.org/10.1007/s00382-019-04704-5)
692 [10.1007/s00382-019-04704-5](https://doi.org/10.1007/s00382-019-04704-5).
- 693 Lindzen, R. S., and B. Farrell, 1980: A simple approximate result for the maximum growth rate of
694 baroclinic instabilities. *Journal of the Atmospheric Sciences*, **37 (7)**, 1648–1654, [https://doi.org/](https://doi.org/10.1175/1520-0469(1980)037(1648:ASARFT)2.0.CO;2)
695 [10.1175/1520-0469\(1980\)037\(1648:ASARFT\)2.0.CO;2](https://doi.org/10.1175/1520-0469(1980)037(1648:ASARFT)2.0.CO;2).
- 696 Merlis, T. M., T. Schneider, S. Bordoni, and I. Eisenman, 2013: Hadley circulation response to
697 orbital precession. part i: Aquaplanets. *Journal of Climate*, **26 (3)**, 740–753, [https://doi.org/](https://doi.org/10.1175/JCLI-D-11-00716.1)
698 [10.1175/JCLI-D-11-00716.1](https://doi.org/10.1175/JCLI-D-11-00716.1).
- 699 Neu, U., M. G. Akperov, N. Bellenbaum, and Coauthors, 2013: IMILAST: A community effort to
700 intercompare extratropical cyclone detection and tracking algorithms. *Bulletin of the American*
701 *Meteorological Society*, **94 (4)**, 529–547, <https://doi.org/10.1175/BAMS-D-11-00154.1>.
- 702 Pinto, J. G., and P. Ludwig, 2020: Extratropical cyclones over the north atlantic and western europe
703 during the last glacial maximum and implications for proxy interpretation. *Climate of the Past*,
704 **16 (2)**, 611–626, <https://doi.org/10.5194/cp-16-611-2020>.

- Priestley, M. D. K., and J. L. Catto, 2022: Future changes in the extratropical storm tracks and cyclone intensity, wind speed, and structure. *Weather and Climate Dynamics*, **3** (1), 337–360, <https://doi.org/10.5194/wcd-3-337-2022>.
- Raavi, P. H., J.-E. Chu, A. Timmermann, S.-S. Lee, and K. J. E. Walsh, 2023: Moisture control of tropical cyclones in high-resolution simulations of paleoclimate and future climate. *Nature Communications*, **14**, 6426, <https://doi.org/10.1038/s41467-023-42033-8>.
- Raddatz, T. J., and Coauthors, 2007: Will the tropical land biosphere dominate the climate–carbon cycle feedback during the twenty-first century? *Climate Dynamics*, **29** (6), 565–574, <https://doi.org/10.1007/s00382-007-0247-8>.
- Raible, C. C., M. Messmer, F. Lehner, T. F. Stocker, and R. Blender, 2018: Extratropical cyclone statistics during the last millennium and the 21st century. *Climate of the Past*, **14** (10), 1499–1514, <https://doi.org/10.5194/cp-14-1499-2018>.
- Raible, C. C., J. G. Pinto, P. Ludwig, and M. Messmer, 2021: A review of past changes in extratropical cyclones in the northern hemisphere and what can be learned for the future. *WIREs Climate Change*, **12** (3), e680, <https://doi.org/10.1002/wcc.680>.
- Shi, X., M. Werner, H. Yang, Q. Gao, J. Liu, and G. Lohmann, 2025: Precession controls on climate and water isotope signals in Northern Africa. *Paleoceanography and Paleoclimatology*, **40**, e2024PA004 999, <https://doi.org/10.1029/2024PA004999>.
- Sidorenko, D., and Coauthors, 2019: Evaluation of FESOM2.0 coupled to ECHAM6.3: preindustrial and HighResMIP simulations. *Journal of Advances in Modeling Earth Systems*, **11** (11), 3794–3815, <https://doi.org/10.1029/2019MS001696>.
- Simmonds, I., and K. Keay, 2000: Mean southern hemisphere extratropical cyclone behavior in the 40-year ncep–ncar reanalysis. *Journal of Climate*, **13** (5), 873–885, [https://doi.org/10.1175/1520-0442\(2000\)013<0873:MSHECB>2.0.CO;2](https://doi.org/10.1175/1520-0442(2000)013<0873:MSHECB>2.0.CO;2).
- Stevens, B., and Coauthors, 2013: Atmospheric component of the MPI-M Earth System Model: ECHAM6. *Journal of Advances in Modeling Earth Systems*, **5** (2), 146–172, <https://doi.org/10.1002/jame.20015>.

- 732 Ulbrich, U., G. C. Leckebusch, and J. G. Pinto, 2009: Extra-tropical cyclones in the present and fu-
733 ture climate: a review. *Theoretical and Applied Climatology*, **96** (1–2), 117–131, [https://doi.org/](https://doi.org/10.1007/s00704-008-0083-8)
734 10.1007/s00704-008-0083-8.
- 735 Varma, V., and Coauthors, 2012: Holocene evolution of the southern hemisphere westerly winds
736 in transient simulations with global climate models. *Climate of the Past*, **8** (2), 391–402,
737 <https://doi.org/10.5194/cp-8-391-2012>.
- 738 Wallace, J. M., G.-H. Lim, and M. L. Blackmon, 1988: Relationship between cyclone tracks,
739 anticyclone tracks and baroclinic waveguides. *Journal of the Atmospheric Sciences*, **45** (3),
740 439–462, [https://doi.org/10.1175/1520-0469\(1988\)045<0439:RBCTAT>2.0.CO;2](https://doi.org/10.1175/1520-0469(1988)045<0439:RBCTAT>2.0.CO;2).
- 741 Wang, Y., and Coauthors, 2008: Millennial- and orbital-scale changes in the east asian
742 monsoon over the past 224,000 years. *Nature*, **451** (7182), 1090–1093, [https://doi.org/](https://doi.org/10.1038/nature06692)
743 10.1038/nature06692.
- 744 Welch, B. L., 1947: The generalization of ‘Student’s’ problem when several different population
745 variances are involved. *Biometrika*, **34** (1–2), 28–35, <https://doi.org/10.1093/biomet/34.1-2.28>.
- 746 Zhong, Y., and Coauthors, 2024: Orbital controls on North Pacific dust flux during the
747 late Quaternary. *Geophysical Research Letters*, **51**, e2023GL106631, [https://doi.org/10.1029/](https://doi.org/10.1029/2023GL106631)
748 2023GL106631.